

APPROVED METHOD:
MEASURING OPTICAL RADIATION
MAINTENANCE OF LED LAMPS, LIGHT
ENGINES, AND LUMINAIRES
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-84-20 ERRATA

If you, as a user of ANSI/IES LM-84-20, believe you have located an error not covered by the following revisions, you should e-mail your information to Pat McGillicuddy, pmcgillicuddy@ies.org, or send a letter to Pat McGillicuddy, Manager of Standards Development, IES, 120 Wall St. 17th Floor, New York, NY 10005.

Please confine your comments to specific typographical errors or misstatements of fact in the document's text and/or graphics. Do not attempt revisions of ANSI/IES LM-84-20.

Additions will be posted to this list online as they become available. This errata list was last updated **April 7, 2021**.

Additions are shown in **bold face**. Deletions are shown as ~~strikeout~~.

In Section 7.8 Reporting DUT Failures:

A DUT shall be considered to have failed if it does not meet the criteria provided in Section 3.5 **3.2**. Checking for DUT failures as defined in Section 3.5 **3.2**, either by visual observation or automatic monitoring, shall be done at every photometric and electrical measurement interval. Failed DUTs shall be reported. Failed DUT photometric and electrical measurement data shall be excluded from the test report (see Section 8.8).

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Publication of this report
has been approved by IES.
Suggestions for revisions
should be directed to IES.

Prepared by:
The Testing Procedures Committee



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Introduction and Scope

1.1 Introduction

The method for measuring luminous flux and color maintenance of LED light sources has been documented in *ANSI/IES LM-80-20, Approved Method: Measuring Lumen Flux of LED Packages, Arrays, and Modules*.¹ For solid-state lighting (SSL) products such as LED lamps, light engines, and luminaires, other components, in addition to the LED light sources, also contribute to changes in the optical output over time. These system performance changes over time can be directly tested at the SSL product level. Some of the performance metrics of interest include luminous flux, photon flux, radiant flux, chromaticity, and spectral power distribution (SPD). This Approved Method addresses the evaluation of the changes in performance of SSL products over time and can be a useful tool for engineering evaluations of flux maintenance and chromaticity shift for entire assemblies when environmental considerations and variability in depreciation of the LED packages used in the assembly are incorporated into the analysis.

Furthermore, system performance of SSL products such as LED integrated lamps, non-integrated lamps, LED light engines, LED luminaires, OLED light engines, and OLED luminaires are typically, but not without exception, affected by operational and environmental variables such as operating cycle and thermal conditions resulting from auxiliary equipment and fixtures, ambient temperature, airflow, and orientation. This test method has been developed to assist a laboratory in achieving reproducible test results and to establish consistent operational and environmental conditions across laboratories to facilitate reliable comparison of results.

SSL products have already been adopted in horticultural and other applications. It is expected that SSL products will be adopted in new lighting applications and penetrate additional markets. As a result of the additional applications for SSL products, the scope of this Approved Method has been expanded to include reporting of both photon flux and radiant flux in addition to luminous flux.

In addition, it is acknowledged that the field of lighting continues to evolve, and as a result, it is expected that new metrics will be developed. With this expectation, the scope of this Approved Method has also been expanded to include reporting of the SPD to facilitate the use of historical data to calculate new metrics as they are developed and adopted.

1.2 Scope

This document provides the method for measuring the luminous, radiant, and/or photon flux, as well as chromaticity and other spectral properties over time. This method applies to LED lamps, LED light engines, LED luminaires, OLED light engines, and OLED luminaires. The method describes the procedures to be followed and the precautions to be observed in both initial determination and reproduction of results for determining luminous, radiant, and/or photon flux, and spectral power distribution (SPD) and chromaticity as spectral maintenance measurements, under standard operating conditions.

This Approved Method does not provide guidance or recommendations regarding predictive estimations or extrapolation of lumen maintenance beyond the measurements made using this method.

2.0 Normative References

2.1 ANSI/IES LM-79-19

Illuminating Engineering Society. Approved Method: Optical and Electrical Measurements of Solid-State Lighting Products. New York: IES; 2019.

2.2 ANSI/IES LS-1-20

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020.

2.3 ANSI/IES TM-27-20

Illuminating Engineering Society. Technical Memorandum: Standard Format for the Electronic Transfer of Spectral Data. New York: IES; 2020.

2.4 ANSI/IES TM-28-20

Illuminating Engineering Society. Technical Memorandum: Projecting Long-Term Luminous Flux Maintenance of LED Lamps and Luminaires. New York: IES; 2020.

2.5 ANSI/IES TM-33-18

Illuminating Engineering Society. Technical Memorandum: Standard Format for the Electronic Transfer of Luminaire Optical Data. New York: IES; 2018.

2.6 ISO/IEC Guide 98-4: 2012

International Standards Organization and International Electrotechnical Commission. Uncertainty of Measurement – Part 4: Role of measurement uncertainty in conformity assessment. Geneva: ISO; 2012. (Also: JCGM 106:2012.)

3.0 Definitions

For definitions of terms other than those listed below, the reader is referred to ANSI/IES LS-1-20 (see **Section 2.2**).

3.1 device under test (DUT)

An LED lamp, integrated; LED lamp, non-integrated; LED light engine; LED luminaire; OLED light engine; or OLED luminaire that is being tested.

3.2 DUT failure

For the purpose of this document: the DUT suffers damage due to mishandling; or its luminous flux, photon flux or radiant flux decreases by 90 percent or more (i.e., decreases to 10 percent of the initial flux value or less).

3.3 luminous flux maintenance

The ratio of luminous flux at any selected elapsed operating time under specified operating conditions to the initial luminous flux under the same operating conditions (typically expressed as a percentage). *Luminous flux maintenance* is the complement of *luminous flux depreciation* (sometimes called *lumen depreciation*). Often referred to as *lumen maintenance*.

3.4 photon flux maintenance

The ratio of photon flux at any selected elapsed operating time under specified operating conditions to the initial photon flux under the same operating conditions (typically expressed as a percentage). For each measurement interval, the same wavelength range shall be used.

3.5 radiant flux maintenance

The remaining radiant flux output (typically expressed as a percentage of the initial radiant flux output) at any selected elapsed operating time. For each measurement interval, the same wavelength range shall be used.

3.6 tolerance interval

The interval of permissible values of a property. For the purpose of this document, measurement results shall lie within an acceptance interval (defined as the *tolerance interval*) reduced by the expanded uncertainty (95% confidence) of the measurement on both upper and lower tolerance interval limits. ANSI/IES LM-79-19 Annex D and ISO/IEC Guide 98-4 provide further information about tolerance intervals (refer to **Section 2**).

Note: The term *tolerance interval* as used in conformity assessment has a different meaning from the same term as it is used in statistics.

4.0 Physical and Environmental Conditions during Operation and Handling

4.1 General

DUTs should be checked and, as necessary, cleaned prior to testing. Manufacturer handling instructions (e.g., regarding electro-static discharge (ESD)) shall be observed. Variation may occur in luminous flux maintenance values of the DUT due to changes in ambient temperature or to air movement due to the DUT's thermal management design. Therefore, environmental conditions, such as thermal interference from HVAC systems or solar loading, should be reduced to levels reasonably expected to minimize influence relative to operation when the conditions are removed.

4.2 Mounting DUTs

The DUT shall be mounted in accordance with manufacturer recommendations.

4.3 Vibration

DUTs should not be subjected to excessive vibration or shock during operation or handling.

4.4 Ambient Temperature during Operation of DUTs

Unless otherwise specified, ambient temperature during operation of the DUT between photometric measurements shall be maintained at $25\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$. The ambient temperature variation of the operating environment shall be monitored with a sufficient number of and appropriately located temperature measurement points. Ambient temperature sensors shall be shielded from direct optical radiation from the DUTs and from thermal interference from any other sources (e. g., heat sink). Testing shall be suspended if the ambient temperature falls outside of the allowed temperature range.

If an ambient temperature other than $25\text{ }^{\circ}\text{C}$ is specified, the ambient temperature tolerance interval shall be $\pm 5\text{ }^{\circ}\text{C}$ and the test conditions shall be included in the test report.

4.5 Humidity

Humidity of the DUT test environment should be controlled within the testing chamber or room during operation. The nominal setpoint and the variation for the controlled environment should be reported.

4.6 Air Movement

The DUT's self-induced airflow should replicate the DUT's installation or application condition and should not be otherwise restricted. All other airflow to the DUT should be minimized.

4.7 Operating Orientation

The DUT shall be maintained in the same position during operation and for photometric measurement (refer to **Section 6.1**). Operating position shall correspond to the manufacturer's recommended position and shall be reported.

5.0 Electrical Conditions

5.1 Input Voltage

A continuous rated input voltage per the manufacturer's specifications shall be applied for operation of the DUT for the duration of testing. The input source shall meet the applicable requirements of **Section 5.2** or **Section 5.3**.

5.2 AC Line Voltage Wave Shape

The AC power supply shall have a sinusoidal voltage wave shape at the prescribed frequency (typically 60 Hz or 50 Hz) such that the total harmonic distortion does not exceed 3% of the fundamental during the operation of the DUT, unless a wave form altering DUT control device is used.

5.3 Line Impedance

Optionally, the impedance of the electrical system powering the DUT may be identified. Use of AC power supplies with low output impedance may be preferable to maintain consistent test condition.

5.4 Line Voltage Regulation

Branch circuit input voltage shall be regulated to no more than 2% of the rated root-mean square (RMS) value. The input voltage to each DUT or driver shall be verified periodically. The elapsed time between verifications shall not exceed 3,000 hours of DUT operation.

5.5 DC Voltage Regulation

The voltage of a DC power supply (instantaneous voltage) applied to the DUT shall be regulated to within $\pm 0.2\%$ under load. The AC voltage component of the DC regulated voltage shall be less than 0.5% of the RMS of the DC regulated voltage. The input voltage to each DUT or driver shall be verified at periods not to exceed 3,000 hours of DUT operation.

5.6 DC Current Regulation

The input current applied to the DUT during continuous operation between optical measurements shall be monitored and regulated to within $\pm 3\%$ of the DUT rated current per manufacturer's specifications.

5.7 Wiring

Wiring of test racks should be in accordance with national, state or provincial, and local electrical

codes, and in accordance with the manufacturer's recommendations for the DUT or system. For non-integrated lamps, the driver shall be grounded per manufacturer specifications.

An inspection of electric contacts, including the DUT holder contacts, and verification of driver operation should be performed each time a DUT is installed.

6.0 Optical Radiation and Thermal Measurement Procedure

6.1 Optical Radiation Measurement

At each measurement interval (refer to **Section 7**), the DUT shall be removed from test racks and measurements of luminous, radiant, or photon flux; electrical power; system efficacy; spectral power distribution (SPD); and chromaticity performed per ANSI/IES LM-79-19 (see **Section 2.1**). After measurement, the DUT shall be replaced on the test rack for the next cycle, as required.

For dimmable or otherwise controllable DUTs, optical measurements shall be performed at the maximum input power condition.

6.2 LED or OLED Source In-Situ Case Temperature Measurement

Recommended procedures for in-situ LED and OLED source case temperature (T_s) measurements are provided in **Annex A**.

7.0 Testing Procedures

7.1 Seasoning or Aging

Seasoning or aging shall not be performed.

7.2 Handling

When handling, transporting, or storing the DUTs, care should be taken to prevent any damage or contamination that may affect the test results. Handling, transporting, and storage shall be performed per the manufacturer's guidance if such exists.

7.3 DUT

7.3.1 DUT Sample Selection. DUT samples tested shall be selected to be sufficiently representative of the overall population of the type of product being tested. All data collected for the DUT samples shall be used for projections. Specially built samples to be used as DUTs shall not be allowed. Selectively excluding data based on performance for making the projections shall not be allowed. Inclusion of extra DUT samples may be useful if some samples are determined to have failed under the guidelines listed in **Section 7.8**. Documentation of all samples tested shall be included in the report.

7.3.2 DUT Tracking. Each individual DUT shall be tracked during operation. DUTs can be identified by markings applied directly to the DUTs, or by labels that are attached to the DUTs during transport, operation, and evaluation, or to the test rack position occupied by the DUT.

The identification method selected should take into account the effect of exposure to light and heat. Possible suitable marking methods and/or materials include durable bar coding, ceramic ink marking, high temperature markers, or any other method that does not degrade or can be periodically renewed for the duration of the test.

Each DUT sample that undergoes a maintenance test shall be tracked throughout the test.

7.4 Operating Cycle

Continuous operation of the DUT is recommended. Luminous flux and chromaticity maintenance data from varying operating cycles cannot be compared without first evaluating whether the lumen and chromaticity maintenance of the DUT are affected by the operating cycle.

7.5 Timekeeping

Accurate recording of elapsed operating time is critical. Elapsed time recording devices shall be connected to the particular test positions and accrue time only while DUTs are operational. Video monitoring, current monitoring, or other means may be used to determine elapsed operating time if sufficient temporal accuracy

is provided. All equipment used for measuring elapsed operating time shall be calibrated in accordance with manufacturer specifications. Total minimum temporal resolution should be within $\pm 0.5\%$.

If a more precise determination of the time at which a DUT stops operating (e.g., for the purpose of calculating Weibull failure distributions), then more-frequent operation checks may be performed. Typical intervals ranging from 0.5% to 1% of rated life have been used for predicting Weibull distributions of older lighting technologies.

7.6 Operating Duration and Measurement Intervals

Initial photometric and chromatic measurement shall be performed prior to the start of the flux and chromaticity maintenance test and recorded as the initial (time = 0) data.

Photometric measurements of the DUT per ANSI/IES LM-79-19 (see **Section 2.1**) shall be made at each measurement interval of the operating duration.

Because the results of the test described herein may be used for evidence of compliance with regulations as requested or for planned data analysis (e.g., predictive luminous flux maintenance modeling or actual luminous flux maintenance validation), the operating duration and measurement intervals shall be based on and consistent with the DUT specified operating life.

7.7 Operation of Dimmable or Controllable DUTs

If the DUT is dimmable or otherwise controllable, the DUT shall be operated at the maximum input power condition for the specific mode of operation.

A DUT may have multiple modes of operation, including variable correlated color temperature (CCT). The mode(s) of operation specified for each measurement shall remain the same throughout the test, and such setting conditions shall be clearly reported.

7.8 Reporting DUT Failures

A DUT shall be considered to have failed if it does not meet the criteria provided in **Section 3.5**. Checking for

DUT failures as defined in **Section 3.5**, either by visual observation or automatic monitoring, shall be done at every photometric and electrical measurement interval. Failed DUTs shall be reported. Failed DUT photometric and electrical measurement data shall be excluded from the test report (see **Section 8.8**).

8.0 Test Report

The report shall list all pertinent data concerning conditions of testing, type of equipment, and types of DUTs. Typical items reported include administrative information, DUT identification information, the physical and ambient conditions of the test, measurement intervals, the duration of the operation, any non-operational DUTs, other miscellaneous information, any deviation from the recommended method, and reported results.

8.1 Administrative Information

Administrative information may include:

- Testing agency identification
- Report issue date
- Test start date
- Test completion date
- Description of test equipment
- Individual(s) performing test
- Individual(s) reviewing and approving test results

8.2 DUT Identification

Information on all samples tested, specific enough to allow test duplication, shall be provided, including:

- Manufacturer's name
- Design (or model) identifier
- DUT identification; e.g., serial number
- Description of DUT under test
- Non-integrated LED driver or power supply (if applicable)
- Date of manufacture of the DUT

8.3 Physical and Ambient Conditions

- Operating position
- Ambient temperature
- Humidity

8.4 Measurement Intervals

The time between optical testing shall be documented as the measurement interval in hours. The test intervals specified in ANSI/IES TM-28-20 (see **Section 2.4**) should be considered.

8.5 Operating Duration

Operating duration from start of tests to the end of the tests shall be reported in hours. The down time for optical testing shall be excluded.

8.6 Non-operational DUTs

Elapsed time in hours at which a non-operational DUT was observed shall be reported (if applicable).

8.7 Other Items

- Special test conditions shall be reported. (Example: minimum light output of an individual DUT before it is considered a failure, if other than specified in this document; any unusual conditions observed)
- Reported L_{70} (in hours) of individual DUTs (if applicable)
- Statement of uncertainties (if required)

8.8 Deviations

Deviations from the test method listed in this document shall be documented, including as applicable:

- List of nonstandard conditions
- Deviation from nominal or specified operating conditions or testing

8.9 Results

Data for electrical, photometric, and chromatic measurements for each DUT at each interval shall be reported, including:

- Luminous, radiant, and/or photon flux
- Luminous, radiant, and/or photon flux maintenance
- Color quantities, including chromaticity coordinates rounded to 5 significant digits; CCT, CRI, R_f , and R_g for white-light products may optionally be reported
- SPD per ANSI/IES TM-27-20 or ANSI/IES TM-33-18 (see **Section 2**)
- Electrical parameters: input voltage, input current, and input power

The significant digits specified in ANSI/IES TM-28-20 (see **Section 2.4**) should be considered for any related projections.

Annex A – Recommendations for Measurement of In-Situ LED Case Temperature, T_s

This Annex provides guidance on approaches to identifying representative in-situ source case temperature (T_s) conditions within an SSL product. The general goal is to determine the T_s maximum value in the product, as this serves as a worst-case bounding condition up to which all LEDs will operate in the SSL product. In most instances, thermocouple (TC) measurements are taken during thermal assessment associated with safety evaluations.

A.1 Basic Assessment

The use of infrared (IR) thermography or other types of non-contact IR thermometers can be used to conduct a basic assessment of the highest-temperature regions of an LED printed circuit board assembly (PCBA). To utilize these methods, it may be necessary to remove one or more layers of optics from the SSL products so that the IR camera or IR thermometer can receive a direct IR image of the PCBA. **Figures A-1** and **A-2** provide images of PCBAs in a multi-PCBA assembly (exposed with optics removed) and the associated IR thermograph.

Based on the readings of the IR thermograph or “point-and-shoot” style of non-contact IR thermometers, the overall relative temperature profiles of the assembly can be documented, and the highest-temperature groups of LEDs, as well as the highest-temperature LEDs within a group, can be identified. These LEDs are candidates for TC attachment to facilitate accurate T_s measurements using TC methods.

A.2 Symmetry Conditions

In many cases, conditions of symmetry can be used to reduce the total number of LEDs that require TCs to obtain a useful representation of the T_s conditions in the SSL product. **Figure A-3** is an example of a typical PCBA

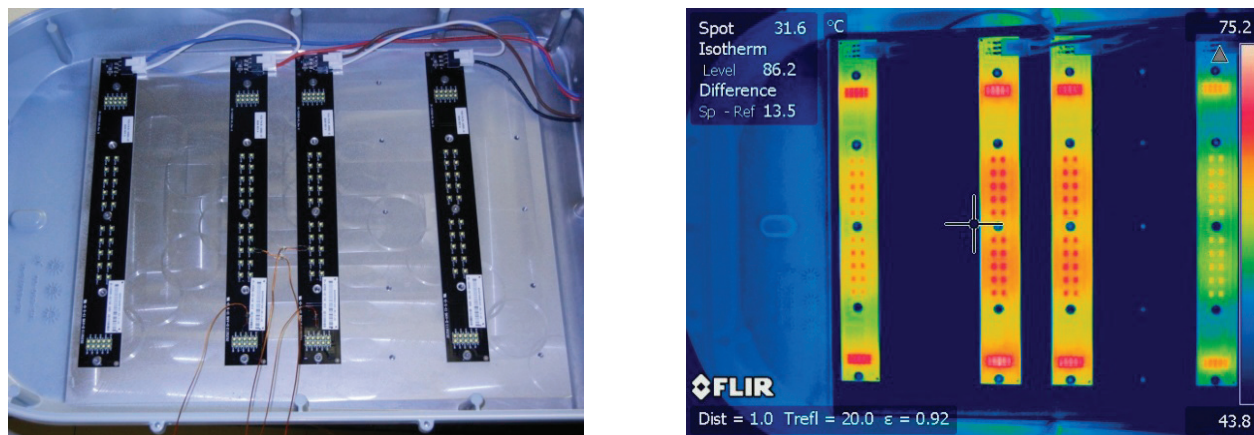


Figure A-1. Left: Typical luminaire assembly with PCBAs exposed for IR thermography or non-contact IR thermometer measurement. Right: Typical IR thermograph from powered PCBAs in a luminaire assembly.

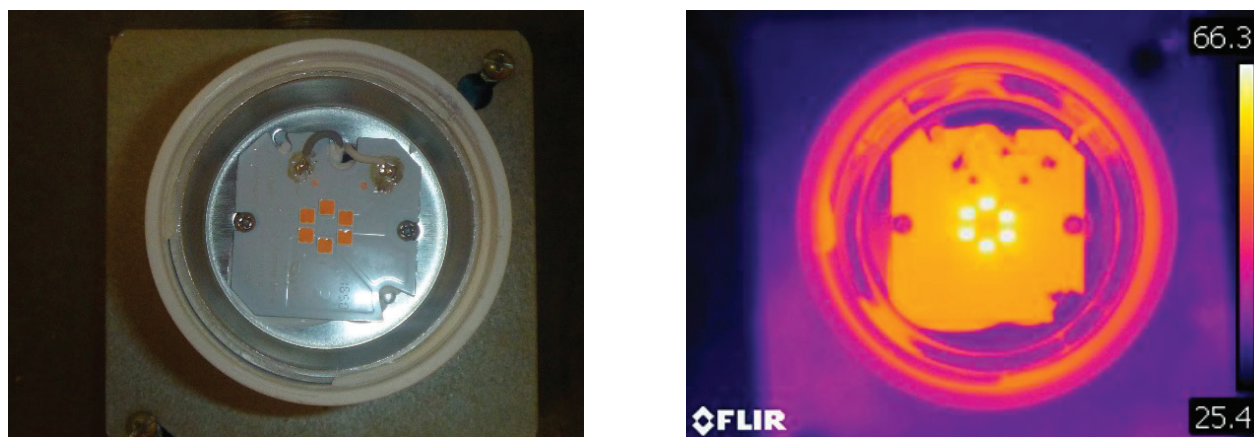


Figure A-2. Left: Typical lamp assembly with PCBAs exposed for IR thermography or non-contact IR thermometer measurement. Right: Typical IR thermograph from powered PCBAs in a lamp assembly.

with mid-line symmetry in two dimensions. In the case of for which symmetry exists, it is sufficient to attach TCs to a subset of the LEDs in reflection of the PCBA symmetry.

A.3 Thermocouples

In all cases using the combined method, LED packages should be instrumented with TCs in accordance with the manufacturer's recommendations for TC location. TCs should be placed in intimate contact with the surfaces indicated in the manufacturer literature and bonded using an appropriate method (e.g., adhesive or soldering) as endorsed by the manufacturer. To ensure that representative values are captured, it is recommended that TCs be attached to multiple LEDs in the assembly.

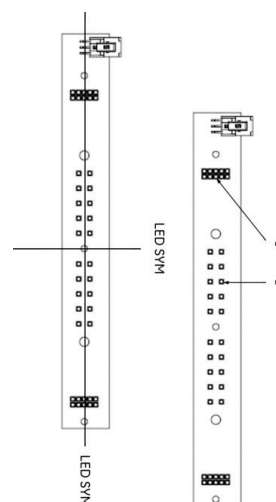


Figure A-3. Left: Example of PCBA with mid-line symmetry. Right: Example of selected LEDs for TC attachment on a PCBA with four-quadrant LED symmetry based on IR thermographic assessment of high-temperature groups.

A.4 Maximum T_s Values

Typically, multiple TCs will be applied to the PCBAs, and multiple PCBAs within an SSL product may be

instrumented. These values should all be recorded and reviewed in order to establish the maximum T_s for the assembly.

ADDITIONAL READING

American National Standards Institute. ANSI C78.377-2017, Electric Lamps – Specifications of the Chromaticity of Solid-State Lighting Products. Washington, DC: ANSI; 2017.

Illuminating Engineering Society. ANSI/IES LM-28-20, Approved Method: Guide for the Selection, Care, and Use of Electrical Instruments in the Photometric Laboratory. New York: IES; 2020.

Illuminating Engineering Society. ANSI/IES LP-4-20, Lighting Practice: Electric Light Sources – Properties, Selection, and Specification. New York: IES; 2020.

Illuminating Engineering Society. ANSI/IES TM-21-19, Technical Memorandum: Projecting Long-Term Lumen Maintenance of LED Light Sources. New York: IES; 2019.

International Commission on Illumination (CIE). CIE Standard S 017/E: 2011 ILV: International Lighting Vocabulary. Vienna: CIE; 2011.

REFERENCES

- 1 Illuminating Engineering Society. ANSI/IES LM-80-20, Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays, and Modules. New York: IES; 2020.



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